

Economic Denial of Sustainability Attack Detection Using Machine Learning

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Abstract—Cloud computing stands as one of the most pervasive technological innovations, providing computing services, including IT infrastructure, over the internet. This technology alone saves many small and medium-sized IT companies from the significant expense of building their IT infrastructure at the beginning of their business journey. Cloud computing offers various user-friendly features, including scalability and flexibility, with a pay-as-you-go model. However, these features come with some drawbacks along with their benefits. Among the challenges posed by the flexibility and pay-as-you-go attributes of cloud computing, one of the most detrimental threats is Economic Denial of Sustainability (EDoS), leading to significant financial losses for consumers of cloud services. Consequently, an effective and efficient solution is urgently needed to fully realize the benefits of cloud computing. In this research, we aim to address the EDoS attack using machine learning-based methods. Our research is conducted on the UNSW-NB15 dataset, and the result demonstrates that our optimized LightGBM model with chosen hyperparameters surpasses existing models with significant improvements in attack detection, achieving a recall score of 99.21%.

Keywords—Economic Denial of Sustainability (EDoS), Cloud Computing, Machine Learning, Deep Learning

I. INTRODUCTION

Since its inception in 1960 [1], cloud computing has undergone rapid evolution and now offers cutting-edge computing services with user-friendly features, making it widely adopted. The recent COVID-19 pandemic has compelled most organizations to operate online, further accelerating the growth of the cloud computing market. According to a report by MarketsandMarkets [2], the global cloud computing market size reached \$545.8 billion by the end of 2020 and is projected to reach \$1240.9 billion by the end of 2027. The primary driver behind the explosive growth of cloud computing is its user-friendly features, which entice companies of all sizes to migrate their IT infrastructure to cloud-based environments. Small and medium-sized startups, in particular, are leveraging cloud computing to establish their IT infrastructure, sparing them from significant upfront investments. Cloud computing comes with many features, such as on-demand self-service, resource pooling, scalability, rapid elasticity, pay-per-use pricing, resilience and availability, broad network access, and so on.

Among these features, pay-as-you-go along with quick elasticity models have especially endeared cloud computing to small and medium-sized companies. Elasticity ensures that computing and networking resources automatically scale up or down based on user demand, and the pay-per-use feature ensures clients only pay for the resources they actually use. The combination of these two features has fueled the popularity of cloud computing. However, these benefits also introduce security concerns. Due to its elastic nature, the cloud allocates additional resources in response to both legitimate and malicious requests. If the requisition is due to illegitimate traffic, the cloud's customer incurs additional charges, leading to financial issues. This malicious exploitation of cloud resources through malicious requests by cybercriminals is termed as an Economic Denial of Sustainability (EDoS) attack [3], posing a significant security challenge to cloud service consumers as well as providers.

EDoS attacks are more challenging to detect than Distributed Denial of Service (DDoS) attacks because they do not render services unavailable but hold up resources unnecessarily to increase the bill. Detecting EDoS attacks efficiently is crucial to fully harnessing the advantages offered by cloud computing technology. Security researchers have investigated this issue and proposed countermeasures, including traditional and machine learning-based approaches. Despite the fact that these methods have significantly improved EDoS attack detection, continued research is necessary to prevent EDoS attacks since they are continually evolving.

Both conventional and deep machine learning techniques have demonstrated high proficiency in various domains, such as computer vision, remote sensing, medical data analysis, and natural language processing. The success and efficiency of machine learning in network traffic flow analysis have inspired us to apply these techniques to the detection of EDoS attacks. We assume that, when trained with optimized hyperparameters, machine learning models can outperform conventional EDoS attack detection methods.

In this paper, our goal is to propose an efficient way to build machine learning models for EDoS attack detection. We will focus on utilizing machine learning models that use less computational resources and take less time to train compared to deep learning models.

Our research highlights the following key contributions:

- Development of a machine learning model that is effective and efficient in EDoS attack detection.

- Finding a machine learning model capable of rapidly predicting the presence of EDoS attacks.

The remainder of this paper is structured as follows: Section II provides studies related to EDoS attack detection. Section III provides the proposed methodology and dataset description. The results and findings obtained are represented and analyzed in Section IV. Section V provides the discussion and future directions for this work. Section VI summarizes our findings and presents the overall conclusions of our work.

II. RELATED WORK

Cloud security, particularly in safeguarding against EDoS attacks, holds paramount significance for cybersecurity researchers, as it stands as a pivotal prerequisite for realizing the genuine benefits of cloud computing. Researchers have been diligently working on EDoS attack prevention since the early stages of its evolution, aiming to shield consumers from the potentially ruinous costs associated with these attacks. Initially, researchers employed various conventional Turing test-based methods in conjunction with firewalls to identify and mitigate EDoS attacks. However, the remarkable success of Artificial Intelligence (AI) and Machine Learning (ML) in traffic flow analysis and other data analysis domains later piqued researchers' interest in utilizing these algorithms for EDoS attack detection.

One notable solution, EDoS-shield [4], introduced the concept of verifier nodes (V-nodes) and virtual firewalls (VFs) for EDoS attack detection and mitigation. V-nodes meticulously scrutinize each request for legitimacy, categorizing them into blacklists or whitelists based on the CAPTCHA response received from the client. Subsequently, the virtual firewall takes action based on these blacklists and whitelists, either blocking or forwarding requests to the cloud accordingly. An enhanced version, Enhanced EDoS-shield [5], improved upon [4] by incorporating Time-to-Live (TTL) field values of IP packets along with IP addresses to update the blacklists and whitelists and determine appropriate actions based on this additional information. In a related approach, as presented in reference [6], a similar methodology to [4] and [5] was employed, but with the inclusion of two additional components: an Intrusion Prevention System (IPS) and a Reverse Proxy (RP) to identify IP packets containing malicious files and to obfuscate the cloud server and load balancer from users. In a proactive stance, the authors in the paper [7] deployed three distinct virtual firewalls at various stages along the path to the cloud to preemptively thwart attacks. To detect any elusive stealth attacks capable of deceiving these three firewalls, they incorporated a performance monitoring system to enhance the overall efficacy of their proposed approach.

In reference [8], different traditional machine learning and deep learning models were harnessed to detect EDoS attacks using the UNSW-NB15 dataset. DLSDN [9] employed a range of models, including Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), CNN-LSTM, SVC-SOM, and SAE-MLP on the dataset [10] for EDoS attack detection. Reference [11] utilized a Stack Deep Polynomial Network (SDPN) on the UNSW-NB15 dataset for detecting EDoS attacks within an SDN-based cloud environment. EDoS-BARRICADE [12] employed a Linear Support Vector Machine (L-SVM) for detecting, isolating,

and mitigating EDoS attacks. R-EDoS [13] introduced Stochastic Recurrent Neural Network (RNN) and Gated Recurrent Unit (GRU) models in conjunction with various feature selection methods for EDoS attack detection. MAN-EDoS [14] harnessed a Multihead Attention Network on the UNSW-NB15 dataset for detecting EDoS attacks.

The papers we have examined employ various conventional as well as machine learning-based methods for the detection of EDoS attacks. Our research concentrates on the utilization of a diverse range of machine learning algorithms for the purpose of EDoS attack detection. We will present a method designed to identify the optimal model configuration and hyperparameters that yield the highest performance.

III. METHODOLOGY

In this research, we propose an approach for detecting EDoS attacks in cloud environments. The approach incorporates various machine-learning techniques for EDoS attack detection using the UNSW-NB15 dataset. A two-phase methodology is employed to develop the final model.

In the initial phase, multiple candidate models are trained using three distinct feature selection methods: Mutual Information, Chi-squared test, and ANOVA. Subsequently, the model demonstrating the highest performance is identified and selected.

In the subsequent stage, the selected top-performing model is fine-tuned with the aim of achieving further improvements in detection accuracy. Figure 1 depicts the details of our proposed methodology.

A. Dataset

Selecting an appropriate and context-relevant dataset is a critical prerequisite for developing machine learning algorithms. In this research, we employed a globally recognized and widely used dataset, named the UNSW-NB15 [15] dataset. This dataset has gained recognition as a benchmark dataset in the network and cloud security fields. The UNSW-NB15 dataset includes data related to nine types of attacks alongside normal traffic. Table I provides a more comprehensive summary of the dataset.

TABLE I
DATASET DETAILS

Category	Count	Percentage
Normal	93,000	46.50%
Generic	58,871	29.43%
Exploits	44,525	22.26%
Fuzzers	24,246	12.12%
DoS	16,353	8.18%
Reconnaissance	13,987	6.99%
Analysis	2,677	1.34%
Backdoor	2,329	1.16%
Shellcode	1,511	0.76%
Worms	174	0.09%
Total	257,673	100%

B. Preprocessing

Effective preprocessing is a crucial step in ensuring the satisfactory performance of machine learning models. The

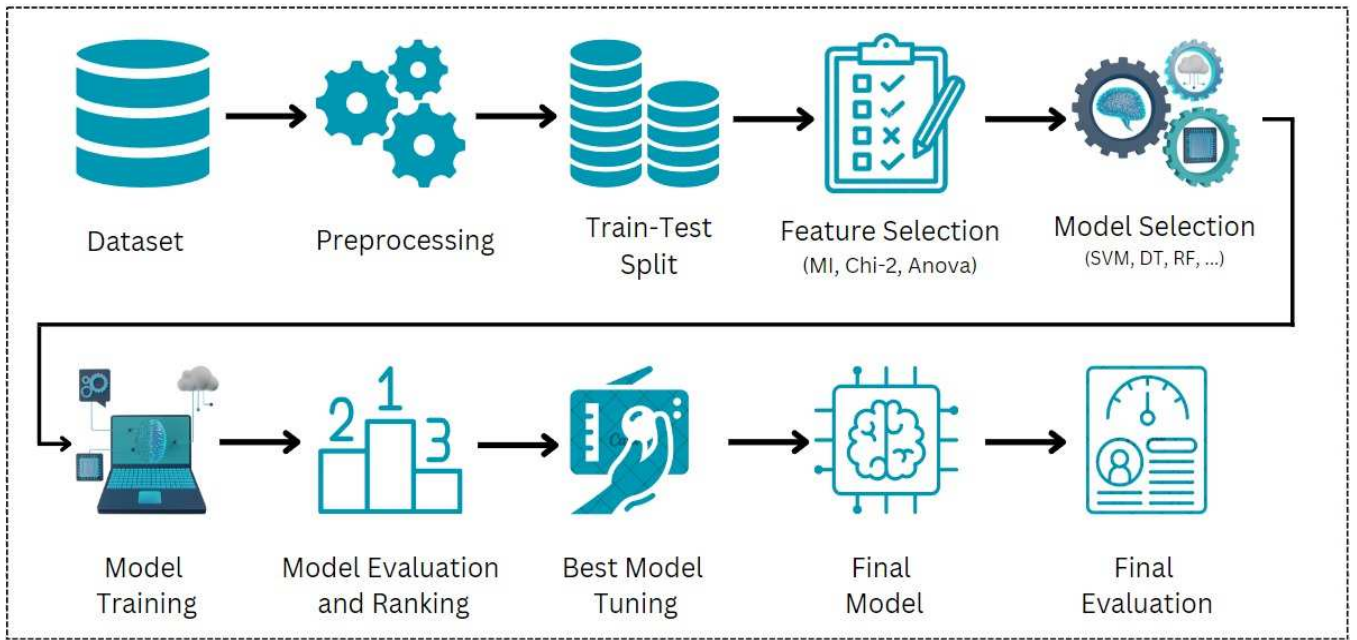


Figure 1. Proposed Methodology.

quality of the data preprocessing significantly impacts the final performance of these models. During the preprocessing stage, the following steps were followed:

1) *Exploratory Data Analysis (EDA)*: EDA was performed to gain a deeper understanding of the dataset's characteristics. EDA provides vital information about the dataset, including the number of features and samples, correlations among different features and target variables, the number of target classes with their distributions, and information regarding missing, outliers, inconsistent, and duplicate data.

2) *Handling Missing and Duplicate Values*: The existence of missing and inconsistent samples within the dataset can adversely affect ML model performance. Therefore, we removed all samples containing missing and duplicate values, as their numbers were negligible compared to the total number of samples in the dataset.

3) *Handling Categorical Variables*: Most traditional machine learning and deep learning algorithms cannot directly process categorical features. Hence, we transformed all categorical values into numerical values using label encoding.

4) *Handling Outliers*: Outliers can also negatively impact ML model performance. But for security-related datasets, removing or transforming outlier values can have a negative impact on the effectiveness of the model. So we kept the outliers unchanged without handling them.

5) *Train-Test Splitting*: To facilitate proper model evaluation and development, we divided our dataset into training and testing sets with a ratio of 80% and 20%, respectively. The training set was used to build the models, and the test set was utilized to evaluate the ML models.

6) *Feature Selection*: Feature selection plays a crucial role in the ML model development process. Extra features

can make a model more complex, which can cause overfitting and lengthen training times. In contrast, selecting appropriate features enhances model performance while reducing the complexity and training time of models. In our study, we applied three feature selection methods, namely Mutual Information (MI), Chi-squared test (Chi-2), and ANOVA.

C. Model Selection

The model selection process comprises two phases. In the initial phase, a variety of fundamental machine learning approaches were chosen, including k-nearest neighbors (KNN), support vector machine (SVM), decision tree (DT), logistic regression (LR), extreme gradient boosting (XGBoost), categorical boosting (CatBoost), random forest (RF), and light gradient boosting machine (LightGBM). This selection encompasses linear models, instance-based models, tree-based models, and ensemble-based models, providing a comprehensive exploration of machine learning algorithms suitable for our dataset. These models were trained and evaluated against the testing data, and subsequent ranking was performed based on their accuracy scores.

In the next phase, we select the top three models with the highest scores for hyperparameter tuning. After fine-tuning, the best performing model was identified as the final choice for EDoS attack detection. Scikit-learn [16] and other Python-based open-source machine learning libraries were employed for the development and testing of those models.

IV. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

In the evaluation of machine learning models, a diverse range of performance metrics are employed, including accuracy, recall, F1-score, precision, Receiver Operating Characteristic (ROC), Area Under the ROC Curve (AUC), kappa score, and more. To provide a comprehensive analysis of the models used in our study, their accuracy, precision, recall, and f1-score are evaluated and presented.

Additionally, the false positive rate (FPR), true positive rate (TPR), AUC score, kappa score, and training-testing time of the best performing model are presented.

We have leveraged several machine learning models, namely Random Forest (RF), Decision Tree (DT), Support Vector Machine (SVM), Logistic Regression (LR), Light Gradient Boosting Machine (LightGBM), K-Nearest Neighbor classifier (KNN), Extreme Gradient Boosting (XGBoost) and Categorical Boosting (CatBoost). The performance metrics of these models for three different feature selection methods (mutual-info, chi-square test, and ANOVA) are presented in Table II, Table III, and Table IV.

TABLE II
PERFORMANCE METRICS OF MACHINE LEARNING MODELS (MUTUAL INFORMATION)

Model	# Opt. Features	Accuracy	Precision	Recall	F1-Score
CatBoost	42	98.98	98.98	98.98	98.98
DT	7	98.71	98.71	98.71	98.71
KNN	7	97.82	97.83	97.82	97.82
LightGBM	33	99.17	99.17	99.17	99.17
LR	42	89.90	90.08	89.90	89.71
RF	4	98.91	98.91	98.91	98.91
SVM	42	92.72	92.79	92.72	92.63
XGBoost	40	98.96	98.97	98.96	98.96

TABLE III
PERFORMANCE METRICS OF MACHINE LEARNING MODELS (CHI-SQUARED TEST)

Model	# Opt. Features	Accuracy	Precision	Recall	F1-Score
CatBoost	39	98.91	98.91	98.91	98.91
DT	32	98.59	98.60	98.59	98.59
KNN	8	96.91	96.93	96.91	96.92
LightGBM	35	99.16	99.16	99.16	99.16
LR	39	90.01	90.19	90.01	89.81
RF	28	98.52	98.52	98.52	98.52
SVM	42	92.72	92.79	92.72	92.63
XGBoost	42	98.88	98.88	98.88	98.88

TABLE IV
PERFORMANCE METRICS OF MACHINE LEARNING MODELS (ANOVA)

Model	# Opt. Features	Accuracy	Precision	Recall	F1-Score
CatBoost	39	98.88	98.98	98.88	98.98
DT	41	98.54	98.54	98.54	98.54
KNN	12	97.48	97.49	97.48	97.49
LightGBM	40	99.15	99.16	99.15	99.15
LR	38	89.84	89.99	89.84	89.64
RF	38	98.43	98.43	98.43	98.43
SVM	42	92.72	92.79	92.72	92.63
XGBoost	33	98.88	98.88	98.88	98.88

These tables show that among the models we used, the LightGBM model with mutual information-based feature selection shows higher performance compared to others with an accuracy score of 99.17%. The precision, recall, and f1-measure of this model are 99.17%, 99.17%, and 99.17% respectively. The number of features used for achieving this score is 33 out of 42 features presented in the dataset. The second and third-best performers with mutual info feature selection are CatBoost and XGBoost, achieving accuracy scores of 98.98% and 98.96%, respectively.

In the cases of chi-squared test and ANOVA feature selection methods, the order of effectiveness remains the same as that of mutual information-based feature selection in terms of accuracy score. However, it's noteworthy that in

both cases, the respective performance scores are lower compared to the mutual information-based feature selection approach.

In the next step, we tuned these three models for further performance increases. After tuning and optimizing hyperparameters, the accuracy score of LightGBM has improved by 0.04% and reached 99.21%. For achieving this score, the LightGBM model was configured with specific parameter values, including 'num iterations' set to 10,000, 'boosting' method as gbdt, 'data sample strategy' using bagging, 'bagging freq' of 1, 'bagging fraction' at 0.9, 'learning rate' of 0.4, 'num leaves' set to 255, and 'max depth' set to 7, while the remaining parameters retained their default values. Tuning and optimizing the hyperparameters of the other two models did lead to performance improvements; however, their performance still does not surpass that of the LightGBM model.

A more detailed classification report of the tuned LightGBM model can be found in Table V. Table VI provides information on the False Positive Rate (FPR), True Positive Rate (TPR), Area under the ROC curve (AUC), kappa score, training time, and testing time of the LightGBM model. It's important to note that this model was trained on 206,138 samples and tested on 51,535 samples. Additionally, Figure 2 displays the confusion matrix, while Figure 3 illustrates the ROC curve of this model.

TABLE V
CLASSIFICATION REPORT OF LIGHTGBM MODEL

Class	Precision	Recall	F1-Score	Support
Normal	98.50	99.32	98.91	18644
Attack	99.61	99.14	99.38	32891
Accuracy			99.21	51535
Macro Avg	99.06	99.23	99.14	51535
Weighted Avg	99.21	99.21	99.21	51535

TABLE VI
FPR, TPR, AUC, KAPPA, TRAINING TIME AND TESTING TIME OF LIGHTGBM MODEL

Class	FPR	TPR	AUC	Kappa score	Training Time	Testing Time
Normal	0.86%	99.32%				
Attack	0.68%	99.14%	99.23%	98.28%	90.91 s	5.92 s

A. Comparison with Existing State-of-the-art Work

Table VII represents a comprehensive comparison of our work with some recently published studies. We present a comparison of studies that utilized the UNSW-NB15 dataset for detecting EDoS attacks. The table illustrates that in the study by Manasa and Joseph (2022) [17], they achieved the highest recall score of 99.00% when compared to other research studies. In contrast, our optimized LightGBM model achieved a recall score of 99.21% showcasing the effectiveness of our model.

V. DISCUSSION AND FUTURE WORK

The EDoS attack is a significant security concern, demanding a robust and efficient mitigation mechanism to ensure the safety and reliability of cloud computing. In response to the security challenges posed by EDoS attacks, this paper

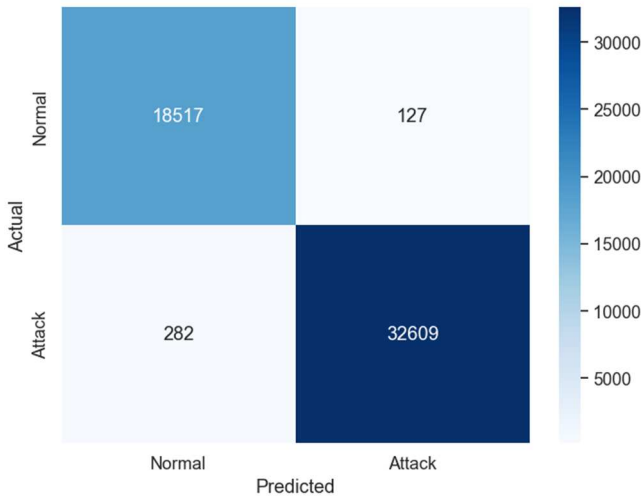


Figure 2: Confusion matrix of LightGBM Model.

presents a machine learning-based approach for their detection and mitigation. Our machine learning models have undergone rigorous training and testing using the UNSW-NB15 dataset. Feature selection techniques and hyperparameter fine-tuning were employed to extract optimal performance from the models.

In our extensive testing of various machine learning models for EDoS attack detection, the LightGBM model has emerged as the standout performer, achieving an outstanding accuracy score of 99.21%. This remarkable performance score surpasses not only the other models used in this study but also the performance of existing state-of-the-art models in this domain.

However, it's essential to note that the method has not been tested in an actual cloud environment. Therefore, we currently lack insight into how our model will perform in a real-world cloud setting. Our future endeavors will involve implementing the method within an experimental cloud environment and evaluating its performance under real-time simulated attacks. This step will provide valuable insights into the practical applicability and effectiveness of our approach.

TABLE VII
COMPARISON WITH OTHER STATE-OF-ART STUDIES.

Ref.	Year	Method	Accuracy	Precision	Recall	F1
Ref. [18]	2021	ML	92.76	96.11	-	94.44
		DL	91.91	94.23	-	93.7
Ref. [14]	2021	DL	98.00	98.90	98.30	98.60
Ref. [11]	2022	DL	96.89	-	97.34	-
Ref. [8]	2022	ML	99.00	98.00	98.00	98.00
		DL	98.27	98.28	94.35	96.28
Ref. [19]	2020	ML	90.85	80.33	98.38	88.45
		DL	86.71	81.54	98.06	89.04
Ref. [17]	2022	ML	99.17	99.00	99.00	99.00
Our Study	-	ML	99.21	99.21	99.21	99.21

VI. CONCLUSION

This research introduces machine learning-based techniques for detecting EDoS attacks. After testing various machine learning models, the LightGBM model emerged as the most

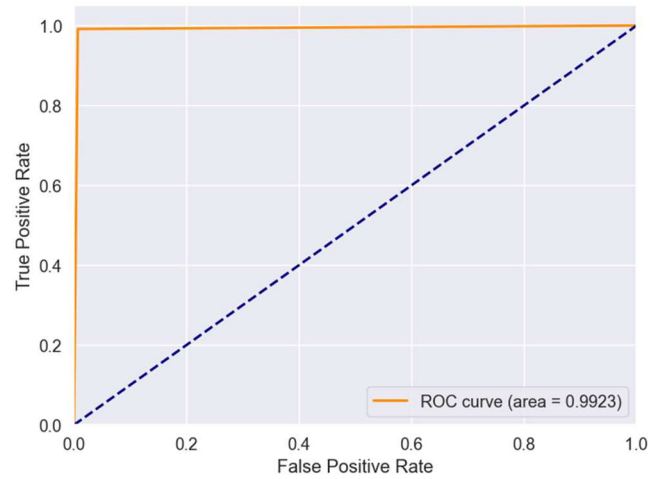


Figure 3: Receiver Operating Characteristic (ROC) curve of LightGBM Model.

effective in EDoS attack detection. With carefully chosen hyperparameters, the LightGBM model surpassed other approaches, achieving an impressive accuracy of 99.21%. These performance results highlight the reliability and effectiveness of the LightGBM model in EDoS attack detection, contributing to the overall improvement of cloud computing security.

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